Explainable Artificial Intelligence (WQF7009)

Assignment 2

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Introduction

Brain tumors represent one of the most severe forms of cancer, requiring rapid and highly accurate diagnosis to determine the appropriate course of treatment. Magnetic Resonance Imaging (MRI) is the gold standard for non-invasive brain tumor detection. However, manual interpretation of these scans is time-consuming and subject to human fatigue or inter-observer variability.

Deep Learning, specifically **Convolutional Neural Networks (CNNs)**, has shown remarkable success in automating the classification of brain tumors into categories such as:

- **Glioma:** Tumors originating in the glial cells.
- **Meningioma:** Typically, benign tumors arising from the membranes surrounding the brain.
- **Pituitary:** Tumors affecting the pituitary gland at the base of the brain.
- **No Tumor:** Normal healthy brain tissue.

Despite their high accuracy, these models are often criticized for their **"black box" nature**. In a medical context, a misclassification could lead to incorrect surgery or treatment. Therefore, it is essential to move beyond simple accuracy metrics and understand *why* a model made a specific prediction.

To bridge this gap, XAI techniques are used:

- **Grad-CAM (Gradient-weighted Class Activation Mapping):** Uses the gradients of the target concept flowing into the final convolutional layer to produce a localization map highlighting the important regions in the image.
- **LRP (Layer-wise Relevance Propagation):** Decomposes the output prediction back through the layers to the input pixels, identifying which specific features contributed most to the final decision.

Objectives

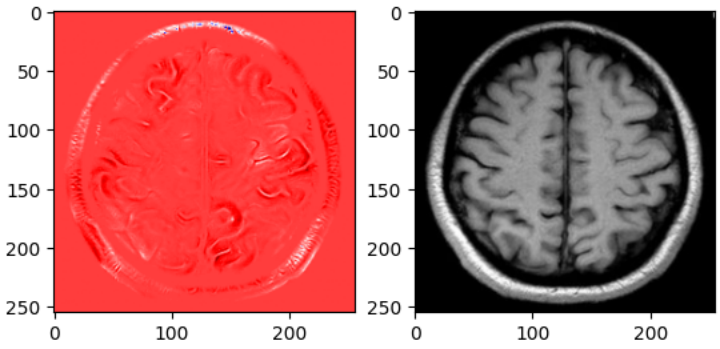
The primary objective of this work is to develop and interpret a robust deep learning framework for the multi-class classification of brain tumors. The specific goals are as follows:

1. Classification Performance: To evaluate the model's ability to accurately distinguish between Glioma, Meningioma, Pituitary tumors, and healthy brain scans using standard performance metrics (Accuracy, Precision, Recall, and F1-score).
2. Visual Interpretability: To implement Grad-CAM and LRP to generate heatmaps that visualize the "regions of interest" (ROI) the model focuses on when making a diagnosis.
3. Use the heatmap to help doctor to explain the diagnosis to patient
4. Use the heatmap to help data scientist explain the diagnosis toa doctor.

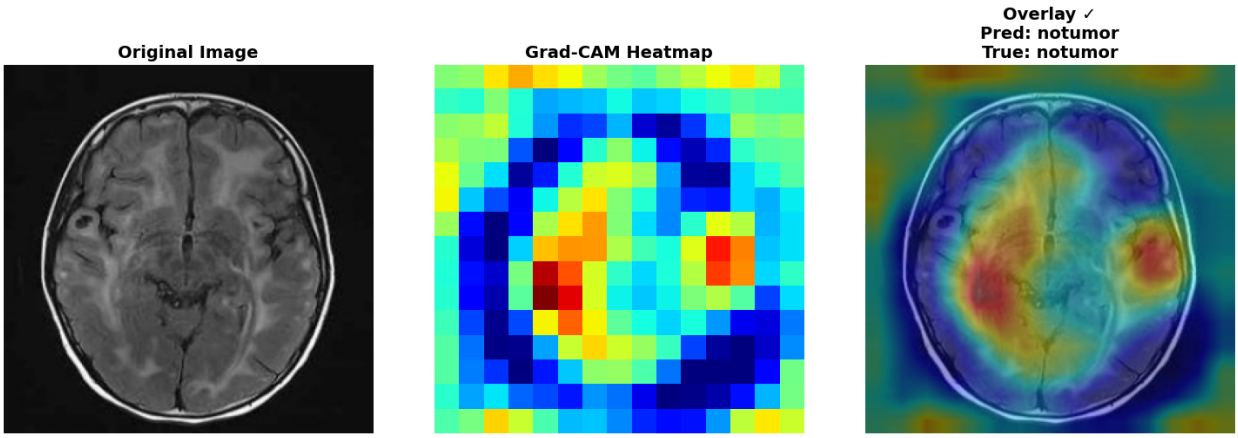
Data Scientist Explains to a Doctor

Feature	Grad-CAM	LRP
Analogy	Like a heat map showing the **general region** of interest.	Like a high-resolution map showing **specific pixel-level** triggers.
Clinical Value	Confirms the model is focused on the lesion and not background noise.	Highlights fine-grained textures or edges that drove the classification.

In the first example, the actual label is no tumor, and the model correctly classified it as no tumor.

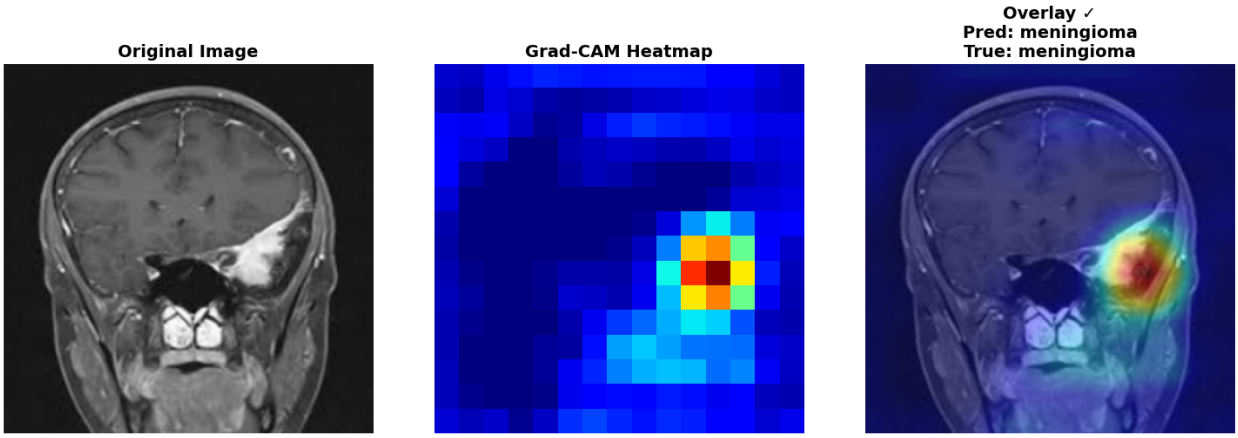


From the LRP image we can see that it did not highlight any pixels.

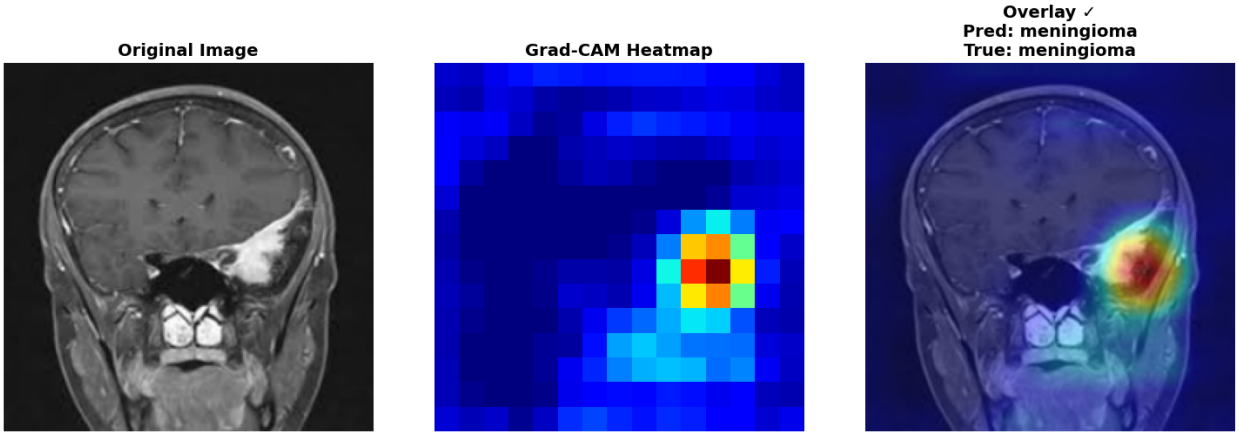


However, Grad-CAM produced a heatmap that is scattered all over the image. In this case LRP offers more interpretability than Grad-CAM.

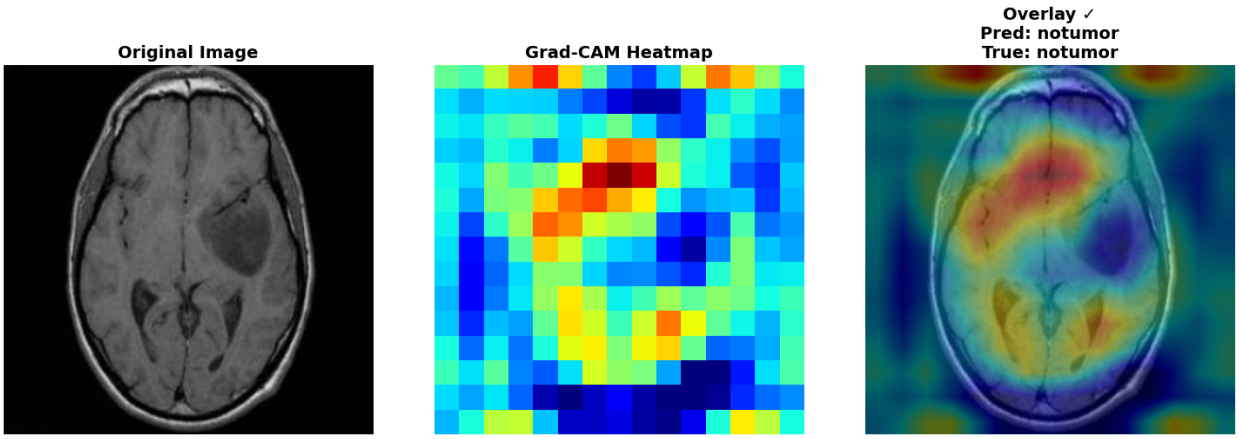
In the second sample the image is labelled as meningioma tumor, and it is correctly classified as meningioma tumor. From here we can clearly observed that the heatmap produced by Grad-CAM clearly highlighted the tumor.



Doctor Explain to a Patient



The AI and I both see a well-defined mass on the outer lining of the brain. These are typically slow-growing and easier to delineate.



The system scanned the entire volume and found no patterns consistent with a growth. It serves as a digital confirmation of a healthy scan.